

FORMELARK

Nomenklatur

Symbol	Beskrivelse	Enhet (SI)
a	Bredde eller annen avstand	m
A	Areal	m^2
b	Høyde, bredde eller annen avstand	m
C_D	Drag koeffisient	
C_L	Løftkoeffisient	
D eller d	Diameter (D er typisk større enn d)	m
$\dot{E}$	Effekt	W
$\dot{E}_{aksling}$	Aksling effekt	W
$\dot{E}_{hyd}$	Hydraulisk effekt	W
f	Friksjonskoeffisient	
$\vec{F}, F$	Kraft	N
F_B	Oppdrift	N
F_D	Dragkraft	N
F_L	Løftkraft	N
$F_{skjær-x}$	Skjærkraft	N
$\vec{g}, g$	Tyngdeakselerasjon, 9,81	m/s^2
h	Høyde	m
h_c	Avstand til geometrisk senter	m
h_l	Tapshøyde	m
$h_p, h_{pump,u}$	Pumpeløfte høyde	m
h_t, h_{turbin}	Tap i turbin	m
HGL	Hydraulic Grade Line	m
$I_{xx, c}$	Areal moment	m^4
K_L	Mindre tap koeffisient	
L, l	Lengde eller avstand	m
L_{typisk}	Typisk lengde, diameter for rør	m
m	Masse	kg
$\dot{m}$	Massestrøm	kg/s
M	Molmasse	kg/kmol, g/mol
$\vec{M}, M$	Moment	$N \cdot m$
P, p	Trykk	N/m^2 ; Pa
p_c	Gjennomsnittstrykk under væskeoverflate	N/m^2 ; Pa
$\vec{r}, r$	Armen til moment, radius	m
Re	Reynoldstall	
t	Tid	s
T	Temperatur	K
$\vec{V}; v$	Hastighet	m/s
V	Volum	m^3
$\dot{V}$	Volumstrøm	m^3/s
y_c	Avstand til geometrisk senter	m
y_p	Avstand til trykksenter	m
z	Avstand opp	m

Greske bokstaver

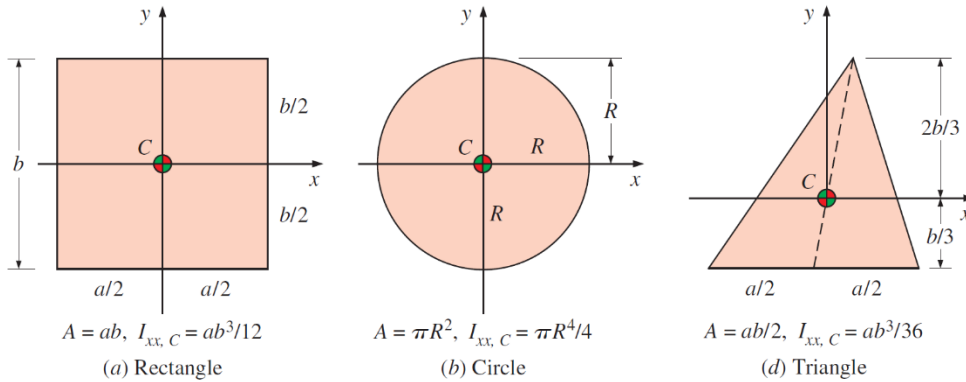
Symbol	Beskrivelse	Enhet (SI)
α	Kinetisk energi korreksjonsfaktor	
ε	Ruhet	m
γ	Spesifikk vekt	N/m ³
η	Virkningsgrad	
μ	Dynamisk viskositet	kg/m·s
ν	Kinematisk viskositet	m ² /s
ρ	Tetthet	kg/m ³
τ	Skjærspenning	N/m ²
ω	Vinkelhastighet	rad/s

Generelt om enheter

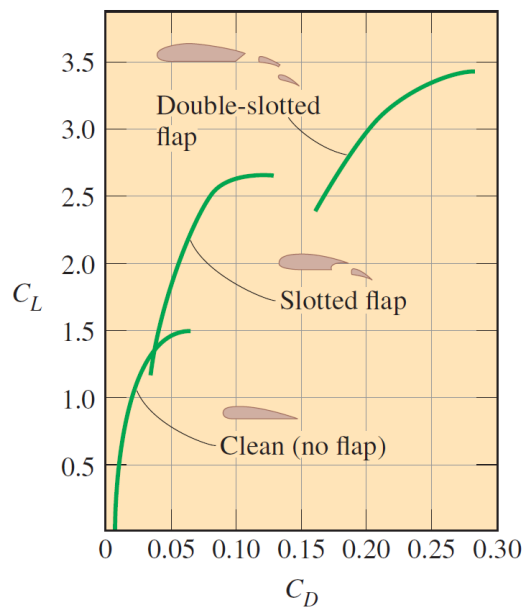
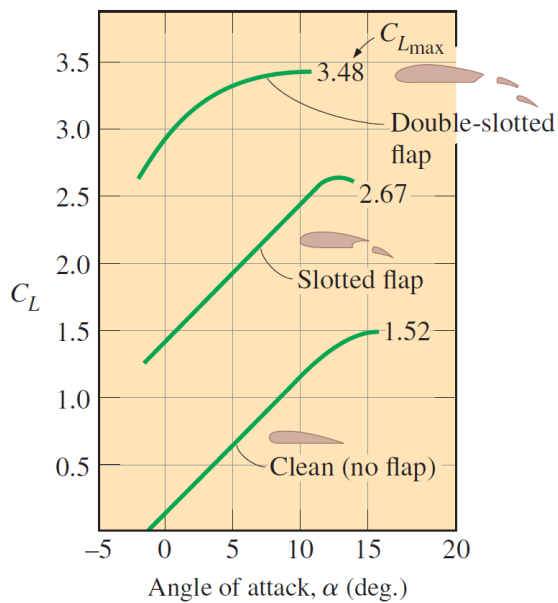
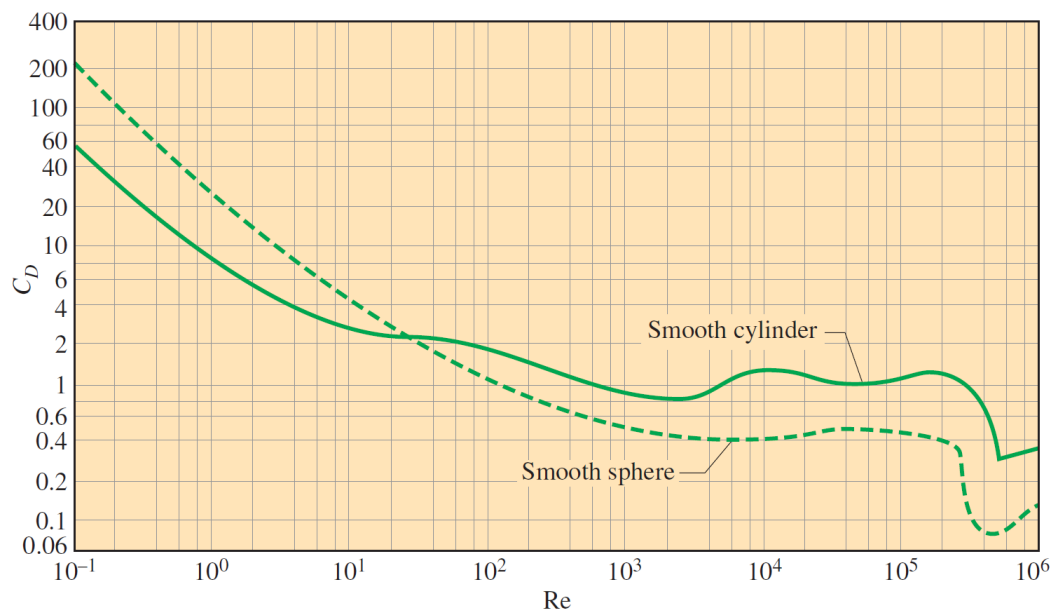
Joule	$J = \frac{kg \cdot m^2}{s^2} = W \cdot s = N \cdot m$
Newton	$N = kg \cdot m / s^2$
Pascal	$Pa = \frac{N}{m^2}$
Watt	$W = \frac{J}{s} = \frac{N \cdot m}{s} = \frac{kg \cdot m^2}{s^3}$

Skjær spenning (N/m ²)	$\tau = \frac{F_{skjær-x}}{A} = \mu \cdot \frac{\partial V_x}{\partial y}$
Hydrostatisk trykk (Pa)	$p = \rho \cdot g \cdot h = \gamma \cdot h$
Gjennomsnitt trykk for overflaten under væske (Pa)	$p_c = \gamma \cdot h_c$
Avstand til trykk senter (m)	$y_p = y_c + \frac{I_{xx,c}}{y_c \cdot A}$
Oppdrift (N)	$F_B = \gamma \cdot V$
Bevaring av masse (kg/s)	$\dot{m}_{inn} - \dot{m}_{ut} = \frac{dm}{dt}$
Uniform massestrøm(kg/s)	$\dot{m} = \rho \cdot v \cdot A = \rho \cdot \dot{V}$
Bevaring av energi (m)	$\frac{p_1}{\rho \cdot g} + z_1 + \alpha \cdot \frac{v_1^2}{2 \cdot g} + h_p = \frac{p_2}{\rho \cdot g} + z_2 + \alpha \cdot \frac{v_2^2}{2 \cdot g} + h_t + h_l$
Bevaring av bevegelsesmengde (N)	$\sum \vec{F} = \sum_{ut} \dot{m} \cdot \vec{V} - \sum_{inn} \dot{m} \cdot \vec{V}$
Friksjonsfaktor for laminær strømning i rør	$f = \frac{64}{Re_D}$
Reynolds tall (-)	$Re = \frac{\rho \cdot v \cdot L_{typisk}}{\mu} = \frac{v \cdot L_{typisk}}{\nu}$
Drag, løft koeffisient (-)	$C_D = \frac{F_D}{\frac{1}{2}\rho v^2 A}, \quad C_L = \frac{F_L}{\frac{1}{2}\rho v^2 A}$
Friksjons tap i rør (m)	$h_l = f \cdot \frac{L}{D} \cdot \frac{v^2}{2g}$
Haaland approximation:	$\frac{1}{\sqrt{f}} \cong -1,8 \times \log \left[\frac{6,9}{Re} + \left(\frac{\varepsilon/D}{3,7} \right)^{1,11} \right]$
Friksjonstap i rørdeler (m)	$h_l = \sum K_L \cdot \frac{v^2}{2g}$
Hydraulisk effekt (W)	$\dot{E}_{hyd} (W) = \Delta p (Pa) \cdot \dot{V} \left(\frac{m^3}{s} \right)$
Aksling effekt (W)	$\dot{E}_{Aksling} (W) = M (Nm) \cdot \omega \left(\frac{rad}{s} \right)$
Virkningsgrad η (%)	$\eta = \frac{Energi\ ut}{Energi\ inn} \times 100$

Geometriske formler for trykksenter



Løft og Drag Koeffisient



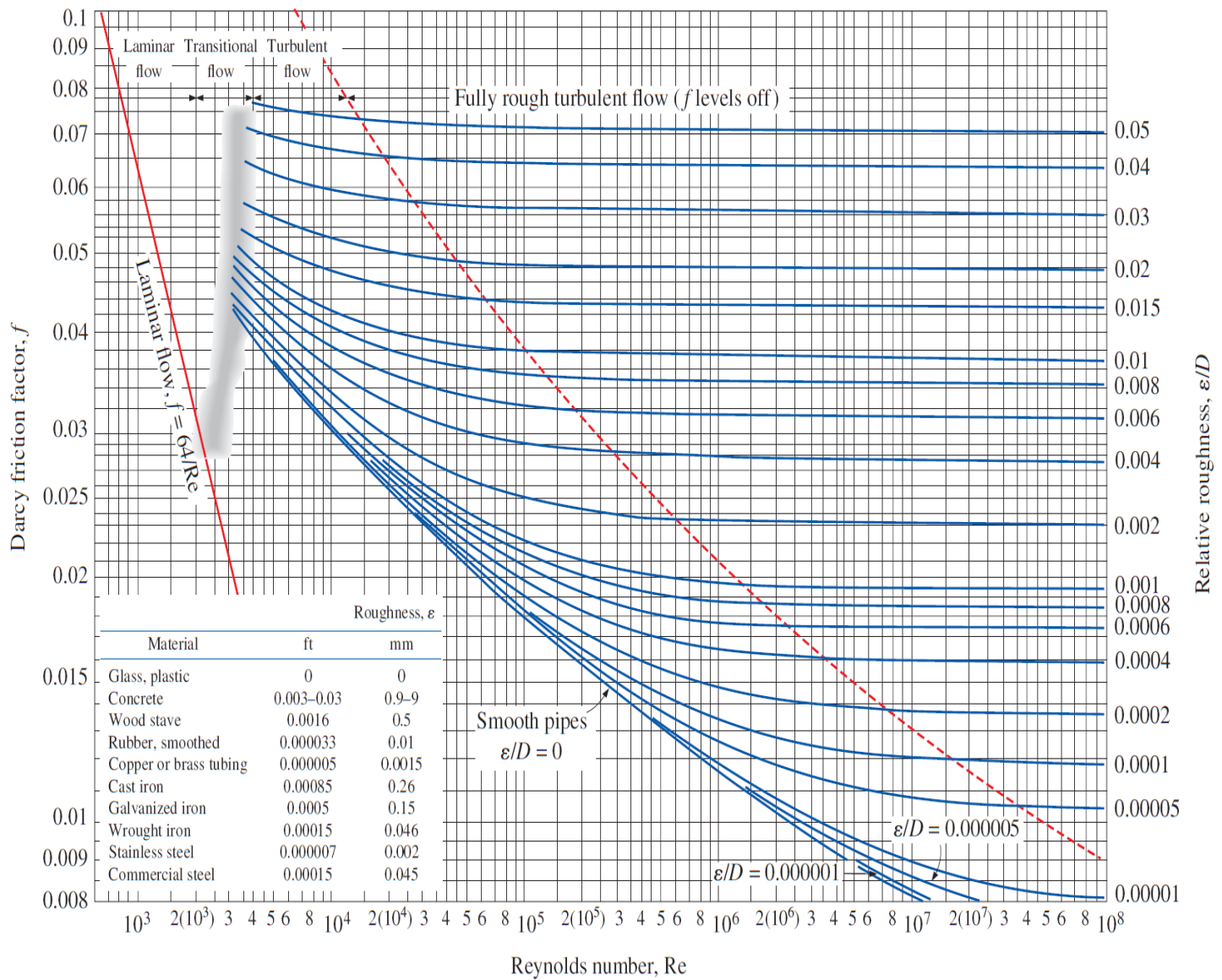


FIGURE A-12

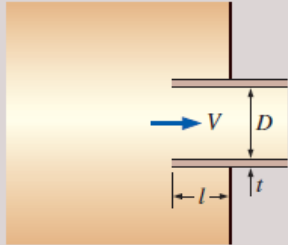
The Moody chart for the friction factor for fully developed flow in circular pipes for use in the head loss relation $h_L = f \frac{L}{D} \frac{V^2}{2g}$. Friction factors in the turbulent flow are evaluated from the Colebrook equation $\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\epsilon/D}{3.7} + \frac{2.51}{Re \sqrt{f}} \right)$.

2 TABELL 8.4.

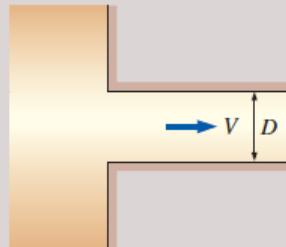
Loss coefficients K_L of various pipe components for turbulent flow (for use in the relation $h_L = K_L V^2 / (2g)$, where V is the average velocity in the pipe that contains the component)*

Pipe Inlet

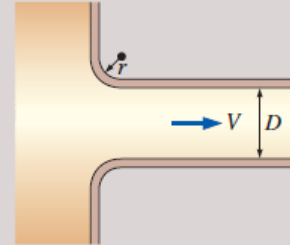
Reentrant: $K_L = 0.80$
($t \ll D$ and $l \approx 0.1D$)



Sharp-edged: $K_L = 0.50$

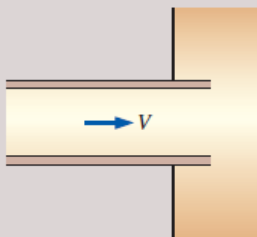


Well-rounded ($r/D > 0.2$): $K_L = 0.03$
Slightly rounded ($r/D = 0.1$): $K_L = 0.12$
(see Fig. 8–40)

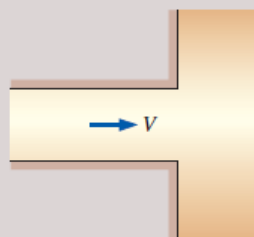


Pipe Exit

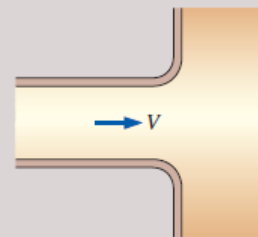
Reentrant: $K_L = \alpha$



Sharp-edged: $K_L = \alpha$



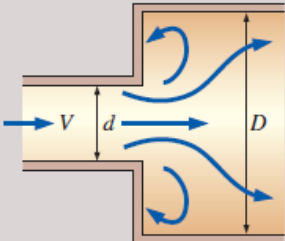
Rounded: $K_L = \alpha$



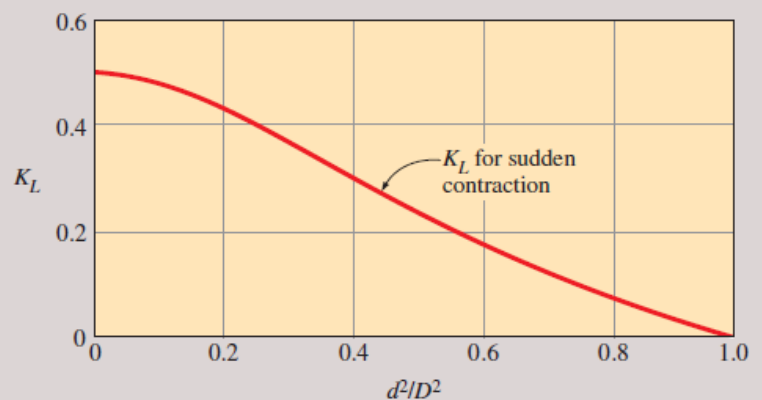
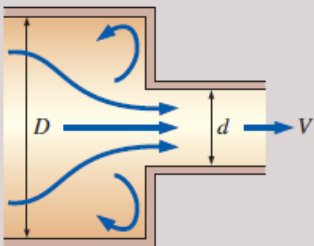
Note: The kinetic energy correction factor is $\alpha = 2$ for fully developed laminar flow, and $\alpha \approx 1.05$ for fully developed turbulent flow.

Sudden Expansion and Contraction (based on the velocity in the smaller-diameter pipe)

Sudden expansion: $K_L = \alpha \left(1 - \frac{d^2}{D^2} \right)^2$



Sudden contraction: See chart.



Gradual Expansion and Contraction (based on the velocity in the smaller-diameter pipe)

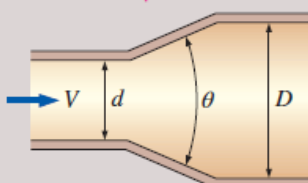
Expansion (for $\theta = 20^\circ$):

$K_L = 0.30$ for $d/D = 0.2$

$K_L = 0.25$ for $d/D = 0.4$

$K_L = 0.15$ for $d/D = 0.6$

$K_L = 0.10$ for $d/D = 0.8$

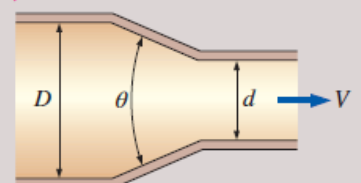


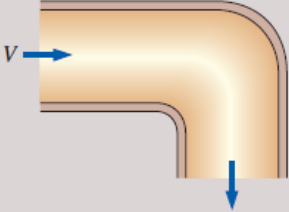
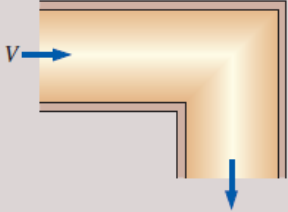
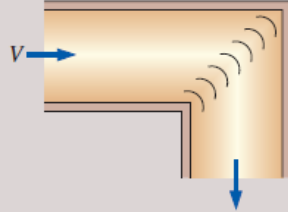
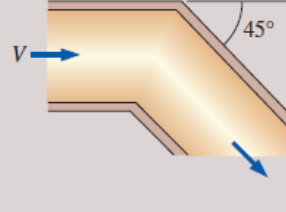
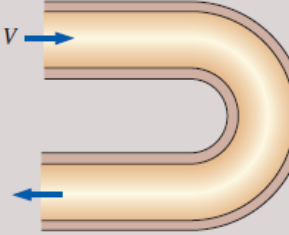
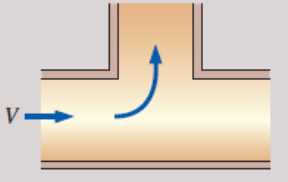
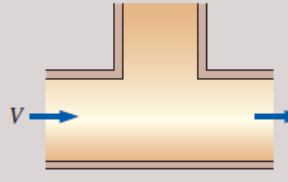
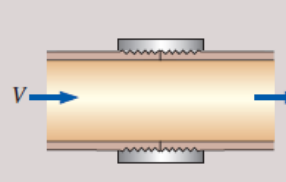
Contraction:

$K_L = 0.02$ for $\theta = 30^\circ$

$K_L = 0.04$ for $\theta = 45^\circ$

$K_L = 0.07$ for $\theta = 60^\circ$



Bends and Branches 90° smooth bend: Flanged: $K_L = 0.3$ Threaded: $K_L = 0.9$ 	90° miter bend (without vanes): $K_L = 1.1$ 	90° miter bend (with vanes): $K_L = 0.2$ 	45° threaded elbow: $K_L = 0.4$ 
180° return bend: Flanged: $K_L = 0.2$ Threaded: $K_L = 1.5$ 	Tee (branch flow): Flanged: $K_L = 1.0$ Threaded: $K_L = 2.0$ 	Tee (line flow): Flanged: $K_L = 0.2$ Threaded: $K_L = 0.9$ 	Threaded union: $K_L = 0.08$ 
Valves Globe valve, fully open: $K_L = 10$ Angle valve, fully open: $K_L = 5$ Ball valve, fully open: $K_L = 0.05$ Swing check valve: $K_L = 2$			
Gate valve, fully open: $K_L = 0.2$ $\frac{1}{4}$ closed: $K_L = 0.3$ $\frac{1}{2}$ closed: $K_L = 2.1$ $\frac{3}{4}$ closed: $K_L = 17$			

* These are representative values for loss coefficients. Actual values strongly depend on the design and manufacture of the components and may differ from the given values considerably (especially for valves). Actual manufacturer's data should be used in the final design.